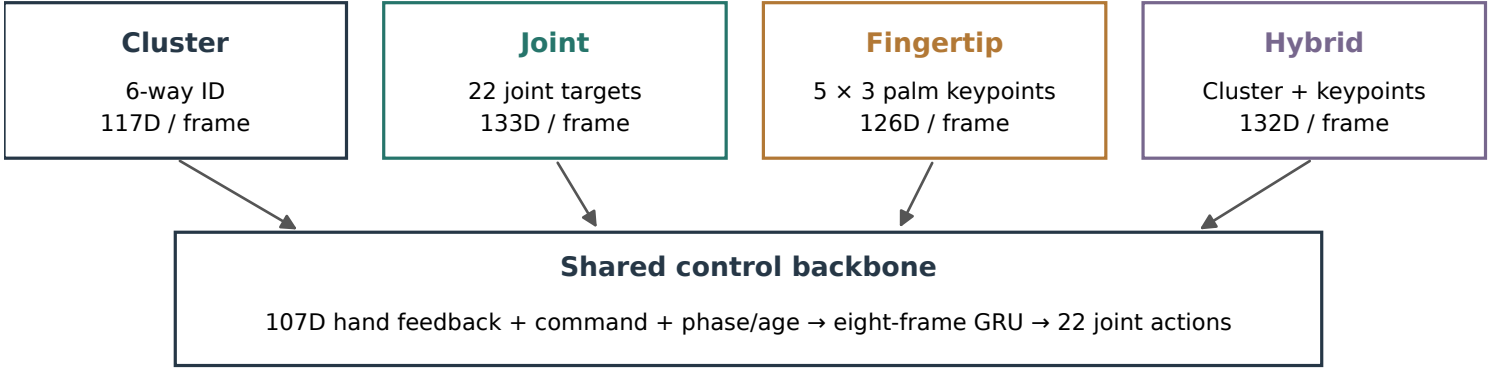
Cluster
6-way ID
117D / frame
Joint
22 joint targets
133D / frame
Fingertip
5 × 3 palm keypoints
126D / frame
Hybrid
Cluster + keypoints
132D / frame
Shared control backbone
107D hand feedback + command + phase/age → eight-frame GRU → 22 joint actions